

SMITHTON, Australia

WMO: 949530

Lat: 40.8347S

Lon: 145.0847E

Elev: 9

StdP: 101.22

Time Zone: 10.00 (AUS)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 0.5	(c) 1.5	(d) -0.3	(e) 3.7	(f) 2.0	(g) 0.6	(h) 4.0	(i) 3.0	(j) 13.5	(k) 10.6	(l) 12.3	(m) 10.3	(n) 2.4	(o) 190	(p) 0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 2	(b) 9.1	(c) 24.1	(d) 18.7	(e) 22.7	(f) 17.9	(g) 21.6	(h) 17.2	(i) 19.6	(j) 22.7	(k) 18.8	(l) 21.5	(m) 18.1	(n) 20.6	(o) 6.8	(p) 70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	13.4	20.9	17.7	12.7	20.5	16.9	12.1	19.9	56.2	22.9	53.6	21.5	51.1	20.6	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.8	11.5	10.5		-1.5	27.8	0.8	2.6	-2.1	29.7	-2.6	31.2	-3.1	32.7	-3.7	34.5	
				-1.6	21.5	0.8	1.1	-2.1	22.2	-2.6	22.9	-3.1	23.5	-3.7	24.3	
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	15.8	16.2	14.7	12.4	10.7	9.2	8.8	9.0	9.9	11.1	13.0	14.2
	DBStd	3.52	2.68	2.74	2.79	2.61	2.38	2.34	2.05	2.08	2.05	2.27	2.48	2.49
	HDD10.0	208	0	0	1	7	21	43	48	44	26	14	3	1
	HDD18.3	2318	87	70	118	177	238	274	294	289	252	225	162	132
	CDD10.0	960	181	174	147	80	41	19	12	14	24	47	92	130
	CDD18.3	28	9	11	5	0	0	0	0	0	0	0	1	3
	CDH23.3	88	41	23	13	1	0	0	0	0	0	0	2	8
	CDH26.7	9	7	1	2	0	0	0	0	0	0	0	0	1
(14) Wind	WSAvg	5.3	6.2	5.9	5.3	4.6	4.5	4.3	4.7	5.0	5.5	5.6	5.8	6.2
(15) Precipitation	PrecAvg	1105	51	46	62	100	116	115	141	129	114	90	77	70
	PrecMax	1452	120	155	158	241	250	214	248	263	210	191	152	228
	PrecMin	799	13	6	7	20	26	42	63	31	17	30	30	1
	PrecStd	182	26	34	37	52	58	39	48	53	49	41	31	48
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.9	25.8	25.0	20.9	17.7	15.6	14.8	15.6	17.2	20.0	22.5	24.6
		MCWB	19.9	19.5	19.4	16.9	14.7	13.3	12.5	12.5	13.5	15.1	17.2	18.8
	2%	DB	24.1	23.8	22.6	19.3	16.6	14.7	13.8	14.4	15.7	17.8	20.5	22.1
		MCWB	18.6	18.8	18.2	16.2	14.4	12.8	11.6	11.8	12.7	13.9	16.3	16.9
	5%	DB	22.4	22.5	21.1	18.3	15.8	14.0	13.1	13.6	14.7	16.6	19.0	20.5
		MCWB	17.7	18.0	17.4	15.6	13.9	12.2	11.1	11.3	12.0	13.4	15.3	16.2
	10%	DB	21.0	21.3	19.9	17.4	15.1	13.3	12.5	12.8	13.9	15.5	17.7	19.2
		MCWB	16.7	17.2	16.5	15.0	13.3	11.7	10.7	10.8	11.5	12.5	14.3	15.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.5	20.6	20.5	18.1	15.7	14.2	13.1	13.2	14.2	16.1	18.4	19.4
		MCDB	24.9	23.8	23.2	19.6	16.8	15.1	14.1	14.7	16.0	17.9	20.9	22.5
	2%	WB	19.4	19.6	19.0	16.9	14.9	13.2	12.2	12.3	13.3	14.9	17.0	18.1
		MCDB	22.9	22.5	21.6	18.4	16.1	14.2	13.2	13.7	14.9	17.1	19.8	20.6
	5%	WB	18.4	18.9	18.0	16.2	14.2	12.6	11.5	11.7	12.6	14.0	15.9	17.0
		MCDB	21.3	21.5	20.4	17.9	15.5	13.6	12.7	13.2	14.2	15.9	18.2	19.5
	10%	WB	17.5	18.1	17.1	15.4	13.6	12.0	11.0	11.2	11.9	13.1	14.8	16.0
		MCDB	20.2	20.7	19.2	17.1	14.9	13.2	12.3	12.6	13.4	15.1	17.3	18.6
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	9.1	9.2	8.8	7.8	7.4	7.0	7.3	7.5	8.4	8.4	8.8
		MCDBR	11.1	9.8	10.0	9.3	7.9	7.0	6.9	7.5	8.2	9.0	9.7	10.4
	5% WB	MCWBR	6.3	5.7	6.1	6.4	6.0	5.6	5.2	5.5	5.7	5.9	6.0	6.3
		MCWBR	9.1	7.8	8.4	7.6	7.0	6.3	6.2	6.9	7.2	8.0	8.1	8.6
(40) Clear-Sky Solar Irradiance	taub	0.352	0.343	0.333	0.331	0.318	0.311	0.311	0.311	0.331	0.366	0.344	0.345	0.342
		taud	2.504	2.547	2.576	2.554	2.579	2.585	2.589	2.524	2.396	2.488	2.503	2.529
	Ebn at Noon	976	960	927	857	803	775	803	844	873	948	977	992	992
		Edh at Noon	111	102	90	80	66	61	65	81	106	107	111	110
(44) All-Sky Solar Radiation	RadAvg	6.68	5.83	4.37	2.98	1.91	1.53	1.66	2.46	3.60	5.10	6.13	6.66	6.66
	RadStd	0.42	0.29	0.21	0.18	0.13	0.13	0.10	0.19	0.22	0.29	0.37	0.45	0.45

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	+0.59	N/A	N/A	N/A	N/A
Regional (16 neighbors)	+0.28	N/A	+0.30	+0.66	+0.40	+0.43	N/A	-86	+91	+10

Nomenclature: See separate page